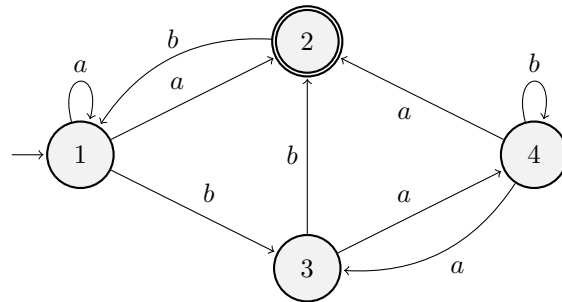


Problem Set 1

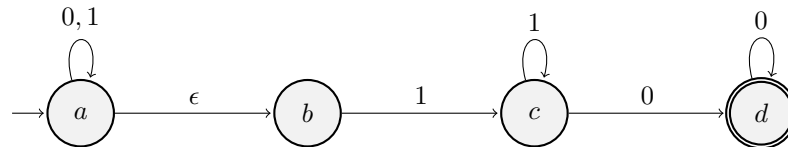
Spring 2024. CSE 309: Theory of Automata

Instructions: This homework is due on **Friday, February 16, 2024**. You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in *zero credit*. An additional key requirement is that you should be able to explain what you submit. Inability to do so will also result again in zero credit. Direct your all queries to the instructors.

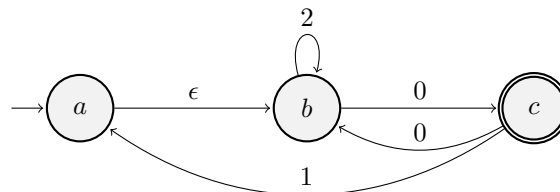
1. Let $S(n) = 1 + 2 + \dots + n$ be the sum of first n natural numbers and let $C(n) = 1^3 + 2^3 + \dots + n^3$ be the sum of first n cubes. Prove that $C(n) = S(n)^2$. Prove the following equalities by induction on n , to arrive at the conclusion that $C(n) = S(n)^2$.
 - (a) (05 points) $S(n) = n(n+1)/2$
 - (b) (05 points) $C(n) = n^2(n+1)^2/4$
2. For each of the following, draw a state diagram of a DFA that recognizes the specified language. In all cases $\Sigma = \{0, 1\}$.
 - (a) (05 points) $L = \{w : w \text{ begins with } 0 \text{ and ends with } 110\}$
 - (b) (05 points) $L = \{w : w \text{ does not contain } 010 \text{ as substring}\}$
 - (c) (05 points) $L = \{w : w \text{ starts with } 1 \text{ and has length multiple of } 5\}$
3. For each of the following, draw a state diagram of an NFA that recognizes the specified language. In all cases $\Sigma = \{0, 1\}$.
 - (a) (05 points) $L = \{(0 \cup 1)^* 00(0 \cup 1)^*\}$
 - (b) (05 points) $L = \{0^{2k} \cup 1^{3k} : k \in \mathbb{N}, k \geq 0\}$
 - (c) (05 points) $L = \{w : w \text{ contains at least two } 0\text{'s and at most one } 1\}$
4. Convert the following NFA's to equivalent DFA's.



(a) (5 points)



(b) (05 points)



5. (05 points) Show that a *finite language is regular*.

6. (05 points) Let L be a regular language. Show that the following language L' is also regular.

$$L' = \{x : \text{there exists } y \text{ such that } xy \in L\}.$$

7. (10 points) We say a state $q \in Q$ is a *dead end* if $\delta^*(q, w) \cap F = \emptyset$ for all $w \in \Sigma^*$. Argue that we can without loss of generality assume that if q is a dead end, then it is also a *trap state*. A trap state is one from which you can never escape. Argue that we can assume that a DFA has at most one trap state. Finally, show that a regular language L is accepted by a DFA with no trap state if and only if $\text{PREFIX}(L) = \Sigma^*$.

[**Note:** $\text{PREFIX}(L)$ is the set of all prefixes of strings in L . For example, if $L = \{\mathbf{ab}, \mathbf{abc}, \mathbf{b}\}$, then $\text{PREFIX}(L) = \{\epsilon, \mathbf{a}, \mathbf{ab}, \mathbf{abc}, \mathbf{b}\}$.]
